Week 3 progressive hints

Use the hints in order. After each hint, return to your own work and continue until you reach a new, specific impasse.

W03-Q01-PA

1. Write $a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{w}$ and compare the three coordinates.
2. The first two coordinate equations determine a and b. Substitute those values into the third coordinate equation before deciding membership.

W03-Q02-PA

1. Look for a direct linear relation among the three displayed vectors before row-reducing.
2. Once one vector is written as a combination of the other two, check that the remaining two are not scalar multiples; this determines the dimension of the span.

W03-Q02-PB

1. Put $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4$ into the columns of a 3×3 matrix. A non-zero determinant establishes independence.
2. For the coordinates, solve $a\mathbf{u}_1 + b\mathbf{u}_2 + c\mathbf{u}_4 = \mathbf{x}$.
3. Connect uniqueness of the coefficient triple to linear independence of the ordered basis vectors.

W03-Q03-PA

1. Form $\mathbf{A} - \lambda\mathbf{I}_2$ and set its determinant equal to zero.
2. Expand the determinant to a quadratic and factor it over the real numbers.

W03-Q03-PB

1. For each eigenvalue, solve $(\mathbf{A} - \lambda\mathbf{I}_2)\mathbf{v} = \mathbf{0}$ and choose one non-zero solution.
2. Put the eigenvectors into the columns of $\mathbf{P}$, and put their eigenvalues into the corresponding diagonal positions of $\mathbf{D}$.
3. Verify $\mathbf{AP} = \mathbf{PD}$. Check $\det(\mathbf{P}) \neq 0$ before using $\mathbf{P}^{-1}$.

W03-Q04-PA

1. The eigenvalues are visible on the diagonal, but the eigenvectors still have to be calculated.
2. If the columns of $\mathbf{P}$ are the ordered eigenvectors, the eigenbasis coordinates of $\mathbf{u}$ are $\mathbf{P}^{-1}\mathbf{u}$.
3. Apply $\mathbf{D}$ to those coordinates before multiplying by $\mathbf{P}$. Then check the result directly from $\mathbf{Bu}$.

W03-Q05-PA

1. The four PMF values must sum to one.
2. Build the CDF by accumulating masses from left to right. Its value changes only at the four support points.
3. The event $1 < X \leq 3$ selects support points rather than an interval of continuous mass.

W03-Q05-PB

1. Calculate $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$ from the PMF, then use $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.
2. For constants a, b, $\mathbb{E}[a + bX] = a + b\mathbb{E}[X]$ and $\mathrm{Var}(a + bX) = b^2\,\mathrm{Var}(X)$.
3. Identify which of these two formulae contains the additive constant.

W03-Q06-PA

1. Use the binomial PMF $\binom{n}{k}p^k(1-p)^{n-k}$, and use a complement for $\mathbb{P}(K \geq 1)$.
2. For the course's geometric convention, $G > 3$ means that the first three trials are failures.

W03-Q06-PB

1. For $N \sim \text{Poisson}(\lambda)$, $p_N(n) = e^{-\lambda}\lambda^n/n!$ on $\{0, 1, 2, ...\}$.
2. A Poisson random variable has mean and variance λ. A binomial random variable has mean np and variance $np(1-p)$.
3. Compare the two means first; equality of means does not force equality of the remaining distributional features.

W03-Q07-PA

1. A continuous uniform density is constant on its support and integrates to one.
2. The CDF is zero below the lower endpoint, linear between the endpoints and one above the upper endpoint.
3. Obtain the requested probability either by integrating the density over the interval or by subtracting two CDF values.

W03-Q07-PB

1. Under the rate parameterisation, $f_T(t) = \lambda e^{-\lambda t}$ for $t \geq 0$.
2. The exponential survival probability comes from integrating from the threshold to infinity. Its mean and variance are $1/\lambda$ and $1/\lambda^2$.
3. Apply the expectation and variance rules for the linear transformation, remembering to square its multiplicative coefficient for the variance.

W03-Q08-PA

1. Correlation divides covariance by the product of the two standard deviations.
2. Expand variances using $\text{Var}(aX + bY) = a^2\,\text{Var}(X) + b^2\,\text{Var}(Y) + 2ab\,\text{Cov}(X, Y)$.
3. Expand $\text{Cov}(2X - Y, X + Y)$ bilinearly, then substitute the three supplied second-moment quantities.

W03-Q09-PA

1. Each I_j is Bernoulli with success probability equal to the marginal probability that draw j is red.
2. Calculate $\mathbb{E}[I_1 I_2] = \mathbb{P}(I_1 = 1, I_2 = 1)$ using the two draws without replacement.
3. Use $\text{Var}(I_1 + I_2) = \text{Var}(I_1) + \text{Var}(I_2) + 2\,\text{Cov}(I_1, I_2)$, then standardise the covariance to find the correlation.

W03-Q10-PA

1. The second equation is twice the first, so it does not supply a new pivot.
2. Use the third equation to relate x and y, then let one of them be a real free parameter and solve for z.

W03-Q10-PB

1. Use the partition (A, A^c) to calculate $\mathbb{P}(B)$.
2. Bayes' rule divides $\mathbb{P}(B \mid A)\mathbb{P}(A)$ by the total probability of B.

W03-Q10-PC

1. Both logarithm arguments must be positive before the quotient rule is used.
2. On the admissible domain, combine the left-hand side into $\log((x-1)/x)$, then use injectivity of the natural logarithm.